8、9、16 作业答案

8.2 质量为 M 的薄片，其质心为 C，$C\xi$，$C\eta$ 是它在薄片平面内的主轴。O 点在薄片平面内，Ox，Oy 相应地与 $C\xi$，$C\eta$ 平行。求证：薄片相对于 Oxy 的惯性积是 Mab，其中 a，b 是 C 点在 Oxy 中的坐标。

如果薄片关于 $C\xi$，$C\eta$ 的主惯量分别是 nMa^2，mMb^2，并且薄片在 O 点有一与 Ox 轴成 45°角的主轴，求证

$$(n-1)a^2=(m-1)b^2.$$

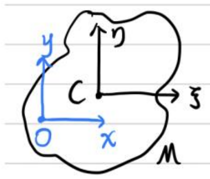
$\because \forall m_i = m_i(\xi,\eta),\ m_i(x,y) = C(x,y)+m_i(\xi,\eta)$

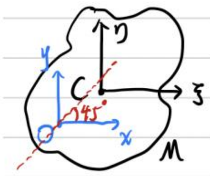
$\therefore \int xy\,dm = \int (a+\xi)(b+\eta)\,dm$

$= Mab + \underline{a\int\eta\,dm} + \underline{b\int\xi\,dm} + \int\xi\eta\,dm$

C是质心

$= Mab$





$\because \int x^2\,dm = \int (a+\xi)^2\,dm = a^2M + \int \xi^2\,dm$

$\int y^2\,dm = \int (b+\eta)^2\,dm = b^2M + \int \eta^2\,dm$

$\therefore I_x = b^2M + nMa^2,\quad I_y = a^2M + mMb^2$

又 $\int xy\,dm = Mab$

$\because I_x\cdot x^2 + I_y\cdot y^2 - 2I_{xy}\cdot xy = k^2$

$M(na^2+b^2)x^2 + M(a^2+mb^2)y^2 - 2Mab\cdot xy = k^2$

令 $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos45^\circ & \sin45^\circ \\ -\sin45^\circ & \cos45^\circ \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$，则 $\begin{cases} x = \frac{1}{\sqrt2}x' - \frac{1}{\sqrt2}y' \\ y = \frac{1}{\sqrt2}x' + \frac{1}{\sqrt2}y' \end{cases}$

$\therefore (na^2+b^2)(x'-y')^2 + (a^2+mb^2)(x'+y')^2 - 2ab(x'-y')(x'+y') = \hat{k}^2$

$\because -(na^2+b^2) + (a^2+mb^2) = 0$

$\therefore (n-1)a^2 = (m-1)b^2$

或直接由椭圆对称性，$I_x = I_y$，即得 $(n-1)a^2=(m-1)b^2$

8.5 均质圆锥体高为 h，底半径为 a，质量为 m。求对于一条母线的转动惯量。

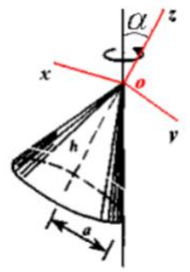
$I_x = \int_M (y^2+z^2)\,dm = \int_V (y^2+z^2)\rho\,dV = \int_V (y^2+z^2)\rho\,dx\,dy\,dz$

令 $x = r\cos\theta,\ y = r\sin\theta,\ z = z$，$\left|\frac{\partial(x,y,z)}{\partial(r,\theta,z)}\right| = r$，$r \le \frac{z}{h}a$

$\therefore I_x = \int_{-h}^{0} dz \int_0^{2\pi} d\theta \int_0^{\frac{za}{h}} (r^2\sin^2\theta + z^2)\rho\, r\, dr$

$= \frac{\rho}{20}a^4h\times\pi + \frac{\rho}{5}\pi a^2h^3$

$\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \frac{\pi}{2}\times\frac{(n-1)!!}{n!!},\ \forall n|2=0$



$\because \rho = \frac{m}{V} = \frac{3m}{\pi a^2h}$

$\therefore I_x = \frac{3}{20}ma^2 + \frac{3}{5}mh^2 = I_y$

又 $I_z = \int_V (x^2+y^2)\rho\,dx\,dy\,dz = \int_{-h}^{0} dz \int_0^{2\pi} d\theta \int_0^{\frac{za}{h}} r^2\rho\, r\,dr$

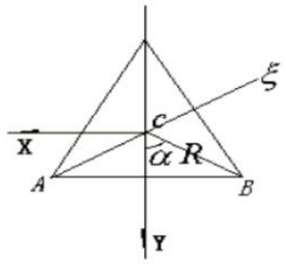
$= \frac{\pi}{10}a^4h\cdot\rho = \frac{3}{10}ma^2$

$\because \overrightarrow{O\xi} = \left(0, \frac{a}{\sqrt{a^2+h^2}}, \frac{h}{\sqrt{a^2+h^2}}\right)$

$\therefore I_\xi = \left(\frac{a}{\sqrt{a^2+h^2}}\right)^2 I_y + \left(\frac{h}{\sqrt{a^2+h^2}}\right)^2 I_z = \frac{ma^2}{20}\times\frac{3a^2+18h^2}{a^2+h^2}$

8.6 解 建立图示cXY坐标系，可以证明X,Y是惯性主轴，且

$$J_X = J_Y$$



∵ y 轴是对称轴，

∴ $J_{XY} = J_{ZY} = 0$ ，又因为 xy 平面是对称面，所以

$J_{XZ} = J_{YZ} = 0$， XY 轴是主轴。再证明$J_X = J_Y$

设J_Y已求得，则由对称性知$J_\xi = J_Y$，在cXY平面内，

$$J_Y = J_\xi = J_X\cos^2\alpha + J_Y\sin^2\beta = J_X\cos^2\alpha + J_Y\sin^2\alpha$$

$$J_Y(1-\sin^2\alpha) = J_X\cos^2\alpha \qquad \Rightarrow \quad J_X = J_Y$$

由此可见，求得J_Y就是过质心任一轴的转动惯量。

计算正 n 边形内任一三角形 ACB 的J_x， J_y

$$J_x = \int_0^{R\cos\alpha} y^2\rho 2x dy = \frac{m}{2n}R^2\cos^2(\frac{\pi}{n})$$

$$J_y = \int_0^{R\cos\alpha}\frac{1}{12}\rho 2x(2x)^2 dy = \frac{m}{6n}R^2\sin^2(\frac{\pi}{n})$$

由平面薄板 $J_z = J_x + J_y$，正 n 边形对过质心 C 的整体J_Z有

$$J_Z = \sum J_z = n(J_x + J_y)$$

$$J_Z = J_X + J_Y = 2J_Y = n(J_x + J_y)$$

$$J_Y = \frac{n}{2}(J_x + J_y) = \frac{n}{2}(\frac{m}{2n}R^2\cos^2(\frac{\pi}{n}) + \frac{m}{6n}R^2\sin^2(\frac{\pi}{n}))$$

即 $J_Y = \frac{m}{12}R^2(2+\cos\frac{2\pi}{n})$

8.9 解 在主坐标下表示角速度，角加速度，动量矩和动能：

$$\vec{\omega} = \omega_x\vec{i} + \omega_y\vec{j} + \omega_z\vec{k}$$

$$\vec{\varepsilon} = \dot{\omega}_x\vec{i} + \dot{\omega}_y\vec{j} + \dot{\omega}_z\vec{k}$$

$$\vec{H}_0 = J_x\omega_x\vec{i} + J_y\omega_y\vec{j} + J_z\omega_z\vec{k}$$

$$T = \frac{1}{2}(J_x\omega_x^2 + J_y\omega_y^2 + J_z\omega_z^2)$$

动量矩矢量和角加速度矢量垂直有：

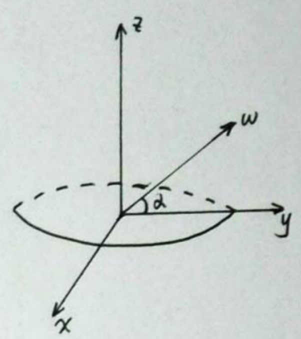
$$\vec{H}_0\cdot\vec{\varepsilon} = 0 = J_x\omega_x\dot{\omega}_x + J_y\omega_y\dot{\omega}_y + J_z\omega_z\dot{\omega}_z$$

动能的变化率：$\frac{dT}{dt} = J_x\omega_x\dot{\omega}_x + J_y\omega_y\dot{\omega}_y + J_z\omega_z\dot{\omega}_z = 0$

即动能 T为常数

8.11. 解. 如右图所示，有

$$\omega_y = \omega\cos\alpha \quad \omega_z = \omega\sin\alpha \quad \omega_x = 0$$



又因为角速度与进动角ψ，章动角θ，自转角φ的关系为

$$\begin{cases} \omega_x = \dot\psi\sin\theta\sin\varphi + \dot\theta\cos\varphi = 0 \\ \omega_y = \dot\psi\sin\theta\cos\varphi - \dot\theta\sin\varphi = \omega\cos\alpha \\ \omega_z = \dot\psi\cos\theta + \dot\varphi = \omega\sin\alpha \end{cases}$$

又由于不受外力矩作用，则 $\omega_x, \omega_y, \omega_z$ 均为常值.

从而可求出 $\theta = \frac{\pi}{2} - \alpha$，$\varphi = 0$ $\quad \Omega = \dot\psi = \omega$

8.12 解由欧拉动力学方程：

$$I_1\dot\omega_\xi + (I_3 - I_1)\omega_\eta\omega_\zeta = -\lambda I_3\omega_\xi \tag{1}$$

$$I_2\dot\omega_\eta + (I_1 - I_3)\omega_\xi\omega_\zeta = -\lambda I_3\omega_\eta \tag{2}$$

$$I_3\dot\omega_\zeta = -\lambda I_3\omega_\zeta \tag{3}$$

由（3）得：$\dfrac{d\omega_\zeta}{\omega_\zeta} = -\lambda dt$ 且 t＝0 时，$\omega_\zeta = \omega_{\zeta 0}$

$$\therefore \omega_\zeta = \omega_{\zeta 0}e^{-\lambda t} \tag{4}$$

将（4）代入（1）（2）得：

$$\frac{d\omega_\xi}{dt} = \frac{I_1 - I_3}{I_1}\omega_\eta\omega_{\zeta 0}e^{-\lambda t} - \lambda\frac{I_3}{I_1}\omega_\xi \tag{5}$$

$$\frac{d\omega_\eta}{dt} = -\frac{I_1 - I_3}{I_1}\omega_\xi\omega_{\zeta 0}e^{-\lambda t} - \lambda\frac{I_3}{I_1}\omega_\eta \tag{6}$$

$$\frac{(5)}{\omega_\eta} + \frac{(6)}{\omega_\xi} \quad \Rightarrow \omega_\xi\frac{d\omega_\xi}{dt} + \omega_\eta\frac{d\omega_\eta}{dt} = -\lambda\frac{I_3}{I_1}(\omega_\xi^2 + \omega_\eta^2)$$

即 $\quad d(\omega_\xi^2 + \omega_\eta^2) = -2\lambda\dfrac{I_3}{I_1}(\omega_\xi^2 + \omega_\eta^2)dt$

$\therefore \quad \omega_\xi^2 + \omega_\eta^2 = Ae^{-\frac{\lambda I_3}{I_1}t} \qquad$ A 为常数

设
$$\begin{aligned} \omega_\xi &= \sqrt{A}e^{-\frac{\lambda I_3}{I_1}t}\sin\varphi(t) \\ \omega_\eta &= \sqrt{A}e^{-\frac{\lambda I_3}{I_1}t}\cos\varphi(t) \end{aligned} \tag{7}$$

（7）代入（5）式化简得：

$$\dot\varphi = \frac{I_3 - I_1}{I_1}\omega_{\zeta 0}e^{-\lambda t}$$

$$\therefore \quad \varphi = -\frac{I_3 - I_1}{\lambda I_1}\omega_{\zeta 0}e^{-\lambda t} + \varepsilon \qquad \text{令} n = \frac{I_3 - I_1}{I_1}\omega_{\zeta 0}$$

则 $\quad \varphi = \dfrac{n}{\lambda}e^{-\lambda t} + \varepsilon \qquad$（$\varepsilon$ 为常数）记 $a = \sqrt{A}$

即
$$\begin{aligned} \omega_\xi &= a\exp\left(-\frac{\lambda I_3}{I_1}t\right)\sin\left(\frac{n}{\lambda}e^{-\lambda t} + \varepsilon\right) \\ \omega_\eta &= a\exp\left(-\frac{\lambda I_3}{I_1}t\right)\cos\left(\frac{n}{\lambda}e^{-\lambda t} + \varepsilon\right) \\ \omega_\zeta &= \omega_{\zeta 0}e^{-\lambda t} \end{aligned}$$

8.14. 解：(1) ∵
$$\begin{cases} I_x\dot{\omega}_x+(I_z-I_y)\omega_y\omega_z=0 \\ I_y\dot{\omega}_y+(I_x-I_z)\omega_z\omega_x=0 \\ I_z\dot{\omega}_z+(I_y-I_x)\omega_x\omega_y=0 \end{cases}$$

将上面三式分别乘以 $\omega_x, \omega_y, \omega_z$，然后相加起来再积分，可知

$$I_x\omega_x^2+I_y\omega_y^2+I_z\omega_z^2=C \quad ①$$

又∵ $\omega_x=\omega l_1 \quad \omega_y=\omega\cdot l_2 \quad \omega_z=\omega\cdot l_3$

l_1, l_2, l_3 均为常量，代入①式，有

$$(I_x l_1^2+I_y l_2^2+I_z l_3^2)\omega^2=C$$

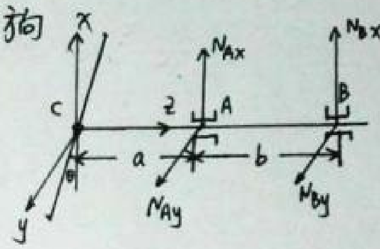
故 ω 为常量

(2) ∵
$$\begin{cases} I_x\dfrac{d\omega_x}{dt}+(I_z-I_y)\omega_y\omega_z=L_x \\ I_y\dfrac{d\omega_y}{dt}+(I_x-I_z)\omega_x\omega_z=L_y \\ I_z\dfrac{d\omega_z}{dt}+(I_y-I_x)\omega_x\omega_y=L_z \end{cases}$$

由于 $\dot{\omega}_x=\dot{\omega}_y=\dot{\omega}_z=0$，则

$$L=\sqrt{L_x^2+L_y^2+L_z^2}=\omega^2\left\{\left[(I_2-I_3)l_2l_3\right]^2+\left[(I_3-I_1)l_3l_1\right]^2+\left[(I_1-I_2)l_1l_2\right]^2\right\}^{1/2}.$$

8.16. 解：如右图所示，以质心C为原点，建立一固结在螺旋桨上的动坐标系Cxyz，令轴Cz与转轴重合，正向为轴承A指向轴承B，轴Cx和Cy所在平面与轴Cz相垂直，A，B两处约束反力为 $N_{Ax}, N_{Ay}, N_{Bx}, N_{By}$，z方向的约束反力为零

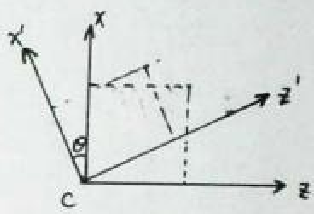


由质心运动定理，可知

$$\begin{cases} -M(\varepsilon y_c + \omega^2 x_c) = N_{Ax} + N_{Bx} - Q \\ -M(\omega^2 y_c - \varepsilon x_c) = N_{Ay} + N_{By} \end{cases}$$

由动量矩定理，可知

$$\begin{cases} I_{yz}\cdot\omega^2 - I_{xc}\cdot\varepsilon = -N_{Ay}\cdot a - N_{By}\cdot(a+b) \\ -I_{xz}\cdot\omega^2 - I_{yz}\cdot\varepsilon = N_{Ax}\cdot a + N_{Bx}\cdot(a+b) \end{cases}$$



由于 $\varepsilon=0$，$x_c=y_c=0$　$I_{yz}=0$

$$\Rightarrow N_{Ax} = Q - \frac{-I_{xz}\cdot\omega^2 - Q\cdot a}{b} \qquad N_{Bx} = \frac{-I_{xz}\cdot\omega^2 - Q\cdot a}{b}$$

$$= Q + \frac{I_{xz}\cdot\omega^2 + Q\cdot a}{b}$$

而 $I_{xz} = \sum m_i x_i z_i$

$$= \sum m_i (z_i'\sin\theta + x_i'\cos\theta)\cdot(z_i'\cos\theta - x_i'\sin\theta)$$

$$= \sum m_i x_i' z_i' \cos 2\theta + \sum m_i (z_i'^2 - x_i'^2)\sin\theta\cos\theta$$

因为轴Cx'，Cz'皆为惯性主轴，则

$$\sum m_i x_i' z_i' \cos 2\theta = I_{x'z'}\cos 2\theta = 0$$

$$\therefore \cancel{I_{x'z'}=}\ I_{xz} = \sum m_i (z_i'^2 - x_i'^2)\sin\theta\cos\theta$$

$$= (I_{x'} - I_{z'})\sin\theta\cos\theta$$

$I_{x'}=0$　　$I_{z'} = \frac{1}{12}ml^2$　　则 $I_{xz} = \frac{1}{12}ml^2\sin\theta\cdot\cos\theta$

$$= \frac{1}{12}\times 7.5\times 1\times \sin(0.015\times 57.32)\times\cos(0.015\times 57.32)$$

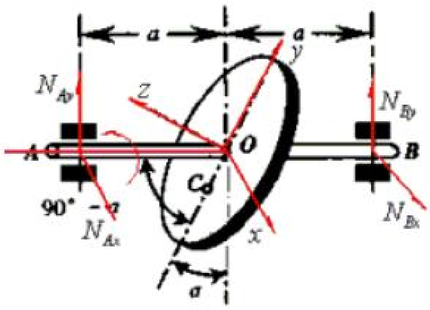
$$= 0.0094$$

$\omega = 2\pi n = 100\pi$ rad/s

$$\therefore N_{Ax} = Q + \frac{I_{xz}\cdot\omega^2 + Q\cdot a}{b} = 75 + \frac{0.0094\times(100\pi)^2 + 75\times 0.1}{0.25} = 3812\,N$$

$N_{Bx} = -3737\,N$.

8.21 **解** 建立图示随体坐标 $oxyz$ 坐标系，重力引起的约束反力为非动约束反力，只分析动约束反力且在随体标架中分解



$$N_{Ax}, N_{Bx}, N_{Ay}, N_{By}$$

x, y, z 都是惯性主轴：

$$I_z = \frac{1}{2}Mr^2 + Me^2,\ I_x = \frac{1}{4}Mr^2 + Me^2,\ I_y = \frac{1}{4}Mr^2$$

$$\omega_x = 0,\ \omega_y = -\omega\sin\alpha,\ \omega_z = \omega\cos\alpha \quad \text{且为常数}$$

代入欧拉动力学微分方程：

$$-(I_y - I_z)\omega_y\omega_z = (N_{By} - N_{Ay})a$$

$$0 = (N_{Ax} - N_{Bx})a\cos\alpha$$

$$0 = (N_{Ax} - N_{Bx})a\sin\alpha$$

由质心动量定理：

$$Me\sin\alpha\omega^2 = N_{Ay} + N_{By}$$

$$N_{Ax} + N_{Bx} = 0$$

可求得：$N_{Ax} = N_{Bx} = 0$

$$N_{By} = \frac{1}{2}Me\sin\alpha\omega^2 + \frac{\omega^2\sin 2\alpha}{16a}M(r^2 + 4e^2)$$

$$N_{Ay} = \frac{1}{2}Me\sin\alpha\omega^2 - \frac{\omega^2\sin 2\alpha}{16a}M(r^2 + 4e^2)$$

16.6 解：对M有：平面的约束反力 $N = Mg$

则 $fMg = M\omega^2 R$

$\Rightarrow R = \frac{fg}{\omega^2}$

物块能保持相对静止的位置在以转轴为圆心、$\frac{fg}{\omega^2}$ 为半径的圆内

16.8 解（1）方法一：由 $m\bar{a}_r = \bar{S}_e + \bar{S}_k + \bar{F}$

$$\bar{F} = \bar{0}$$

将上式在 $\overline{v_r}$ 方向上投影得

$$m\frac{dv_r}{dt} = S_e\cos\theta \qquad \because (\bar{S}_k \perp \bar{v}_r)$$

又 $\overline{v_r} = \dot{r}\overline{e_r} + r\dot{\varphi}\overline{e_\varphi}$

∴ $\cos\theta = \dfrac{\dot{r}}{v_r}$

$$m\frac{dv_r}{dt} = \frac{\dot{r}}{v_r}\cdot mr\omega^2$$

积分得 $\dfrac{1}{2}v_r^2 = \dfrac{1}{2}\omega^2 r^2 + c$

$$v_r^2 - \omega^2 r^2 = 常量$$

即 $\dot{x}^2 + \dot{y}^2 - \omega^2(x^2 + y^2) = 常量$

(1)方法二：由科氏惯性力不作功，牵连惯性力可看成有势力，势函数为：

$$V = -\frac{1}{2}m\omega^2(x^2 + y^2)$$

非惯性系中机械能守恒：

$$\frac{1}{2}mv^2 - \frac{1}{2}m\omega^2(x^2 + y^2) = E$$

即 $\dot{x}^2 + \dot{y}^2 - \omega^2(x^2 + y^2) = 常量$

（2）方法一：由非惯性系中的动量矩定理

$$\frac{d}{dt}(m\bar{r}\times\overline{v_r}) = \bar{r}\times\overline{S_e} + \bar{r}\times\overline{S_k} = \bar{r}\times\overline{S_k}$$

$$S_k = -2m\omega v_r$$

$$\bar{r}\times\overline{S_k} = 2m\omega r v_r\cos\theta\bar{k} = -2m\omega r\dot{r}\bar{k}$$

∴ $\dfrac{d}{dt}(\bar{r}\times\overline{v_r}) = -2\omega\dot{r}r\bar{k}$

$$d(\bar{r}\times\overline{v_r}) = -2\omega r dr\bar{k}$$

$$\bar{r} = x\bar{i} + y\bar{j};\quad \overline{v_r} = \dot{x}\bar{i} + \dot{y}\bar{j}$$

∴ $\bar{r}\times\overline{v_r} = -(y\dot{x} - x\dot{y})\bar{k}$

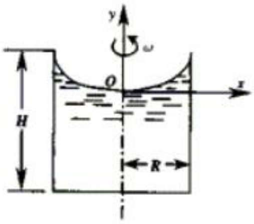
∴ $d(y\dot{x} - x\dot{y} - \omega r^2) = 0$

$$y\dot{x} - x\dot{y} - \omega(x^2 + y^2) = 常数$$

（2）方法二：直角坐标直接投影方程积分。。。。。。

16.12 **证明：**

（1）取动坐标系 oxy 为参考系，取水面上一质点 m 分析



由于 $v_r = 0$ $\therefore$ $a_r = 0$ $S_k = 0$

$$\therefore \qquad o = \vec{S}_e + \vec{p} + m\vec{g}$$

(假设其他质点对其作用力垂直于表面)

于是有 $S_e \cos\theta = mg\sin\theta$， $S_e = m\omega^2 x$

$$\therefore \qquad \mathrm{tg}\theta = \frac{x\omega^2}{g} = \frac{dy}{dx}$$

积分得 $y = \frac{1}{2}x^2\omega^2 / g + c$

解：$x = 0$ 时 $y = 0$ $\therefore c = 0$

$$y = \frac{1}{2}x^2\omega^2 / g \quad \text{为所求曲线}$$

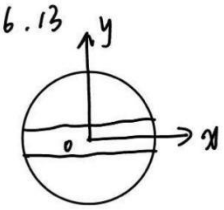
（2） 令 $x = R$，则 $y = \frac{1}{2}R^2\omega^2 / g$

设 注入液体最大高度 H' ，则

$$\pi R^2 H - \pi R^2 H' = \int_0^y dy \cdot \pi x^2$$

$$H' = H - \frac{R^2\omega^2}{4g}$$

16.13



建系 沿质点位移 $x(t)$

任一时刻 $F = m\omega^2 x(t)$

科氏力 $F_k = -2m\omega \cdot v$

$m\ddot{x} = m\omega^2 x(t)$

$\Rightarrow \ddot{x} = \omega^2 x(t)$

$x(0) = a,\ \dot{x}(0) = 0$

解得 $x(t) = a \cdot \cosh(\omega t)$

$N - P - 2m\omega \dot{x} = 0$

$\dot{x} = a\omega \sinh(\omega t)$

$N = P + 2m\omega^2 a \sinh(\omega t)$

$= P + 2\frac{P}{g}\omega^2 a \sinh(\omega t)$

16.22 $S_k = 2m\omega v_r$

$S_e = m\omega^2 r$

$m\ddot{x}_r = m\omega^2 \cdot x_r$

$\Rightarrow \ddot{x}_r = \omega^2 x_r$

$x_r(0) = a,\ \dot{x}_r(0) = v_{r1}$

$\Rightarrow x_r(t) = \frac{1}{2\omega}\left[(a\omega + v_{r1})e^{\omega t} + (a\omega - v_{r1})e^{-\omega t}\right]$

$v_r = \dot{x}_r = \frac{1}{2}\left[(a\omega + v_{r1})e^{\omega t} - (a\omega - v_{r1})e^{-\omega t}\right]$

由 $F_N - F_{IC} + F_{Ie} \cdot \cos\theta = 0$

$\Rightarrow F_N = m\omega\left[(a\omega + v_{r1})e^{\omega t} - (a\omega - v_{r1})e^{-\omega t}\right] + m\omega^2 \cdot \sqrt{R_1^2 - a^2}$

令 $x_r(t) = l + a$

$\Rightarrow t = \frac{1}{\omega}\ln\frac{\omega(a+l) + v_{r2}}{a\omega + v_{r1}}$

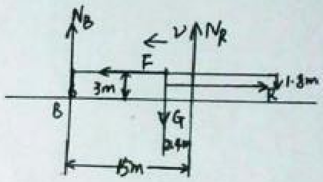
9.7 解. 由条件知 $v_1=200\,km/h$　$v_2=50\,km/h$

∵ $v_2^2-v_1^2=2aS$

$\Rightarrow a=-0.06\,m/s^2$

则惯性力 $F=ma=7500\,N$　方向水平向左

对飞机进行受力分析，如右图所示.

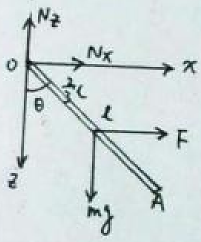


对质心求矩，则有

$$\begin{cases} N_B+N_R=G \\ N_B\times12.6=N_R\times2.4 \end{cases}$$

$\Rightarrow N_B=2\times10^5\,N$

9.12. 解：如右图所示，对杆OA进行受力分析.



惯性力作用在距O点 $\frac{2}{3}l$ 处

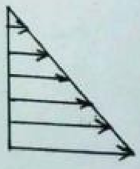
$F=\frac{1}{2}l\sin\theta m\omega^2$

对O点取矩

$mg\cdot\frac{l}{2}\sin\theta=F\cdot\frac{2}{3}l\cos\theta$

$\Rightarrow \cos\theta=\dfrac{3g}{2\omega^2 l}$　$\Rightarrow \theta=\arccos\dfrac{3g}{2\omega^2 l}$

当 $\theta=0$ 时，ω 最小，为 $\omega_{min}=\sqrt{\dfrac{3g}{2l}}$



9.17. 解：由平面运动刚体运动上惯性力简化可知.

$$\begin{cases} ma_x=T\sin33^\circ \\ ma_y=T\cos33^\circ-G \\ a=\sqrt{a_x^2+a_y^2} \end{cases} \Rightarrow \begin{cases} a_x=7.26\ m/s^2 \\ a_y=1.38\ m/s^2 \end{cases}$$

$-J_C\varepsilon+3T\sin3^\circ=0$　$\Rightarrow \varepsilon=\dfrac{628}{J_C}\ rad/s^2$

9.16 对Q分析: $m_Q g = T + m_Q a_Q$

设圆柱中心加速度为 a_C

则顶端加速度 $a_1 = a_C + r\cdot\varepsilon$

$a_1 = a_Q$

对C分析: $T = f + m_C\cdot a_C$

$(T+f)\cdot r = J_C\cdot\varepsilon$

$J_C = \frac{1}{2}m_C r^2$

解得
$$\varepsilon = \frac{m_Q g}{(\frac{3}{4}m_C + 2m_Q)\,r} = \frac{2g}{7r}$$

9.18 解:
$$a_{平动} = \frac{F-f}{m} = \frac{11}{3}\ m/s^2$$

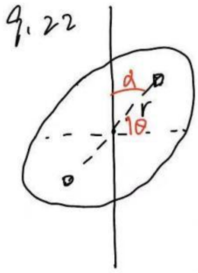
对A点取矩

$$I\cdot\varepsilon = (\frac{1}{3}mL^2)\cdot\varepsilon = F\cdot 0.9m$$

$$\Rightarrow \varepsilon = \frac{25}{18}$$

$$a_C = a + \varepsilon\cdot L = \frac{11}{3} + \frac{25}{18}\times 3.6 = \frac{11}{3} + 5 = \frac{26}{3}\ m/s^2$$

9.22



在圆盘上取两对称小区域

$$dm = r\,d\theta \cdot dr \cdot \rho$$

$$\rho = \frac{m}{\pi R^2}$$

$$dF = dm \cdot \omega^2 r \cos\alpha$$

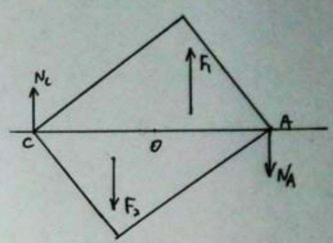
$$dM = r \times dF = r \cdot dm \cdot \omega^2 r \cos\alpha \cdot \sin\alpha$$
$$= r^3 \omega^2 \cos\alpha \cdot \sin\alpha \cdot \rho\, d\theta\, dr$$

$$\therefore M = \int_0^{\pi}\int_0^{R} \rho \cdot r^3 \cdot \omega^2 \frac{1}{2}\sin 2\alpha\, d\theta\, dr$$
$$= \frac{1}{8} m\omega^2 R^2 \sin 2\alpha.$$

$$\because \begin{cases} N_A = N_B \\ N_A \cdot a + N_B \cdot b = M \end{cases} \Rightarrow N_A = N_B = \frac{m\omega^2 R^2 \cdot \sin 2\alpha}{8(a+b)}$$

9.24. 解：由定义可求出该薄板绕AC转动的转动惯量

$$J = \frac{ma^2b^2}{6(a^2+b^2)}$$



则惯性力矩 $L = J\varepsilon = \frac{ma^2b^2\varepsilon}{6(a^2+b^2)}$

该力矩与M同向，则有

$$M = J\varepsilon$$

$$\Rightarrow \varepsilon = \frac{6M(a^2+b^2)}{ma^2b^2\varepsilon}$$

该薄板在A.C处受到N_A，N_c两个约束力，F_1和F_2为两部分转动所产生的离心力，这两组力应满足平衡条件

薄板离心力F_1作用点距C点 $l_1 = \left(\frac{b^4}{2\sqrt{a^2+b^2}}\right)^{\frac{1}{3}}$

$$F_1 = F_2 = \frac{abm\omega^2}{12\sqrt{a^2+b^2}}$$

对o点取矩

$$F_1 \cdot 2\left(l - \frac{\sqrt{a^2+b^2}}{2}\right) = N_A \cdot \frac{\sqrt{a^2+b^2}}{2} \times 2$$

$$\Rightarrow N_A = N_C = \frac{abm\omega^2}{6(a^2+b^2)}\left[\left(\frac{b^4}{2\sqrt{a^2+b^2}}\right)^{\frac{1}{3}} - \frac{\sqrt{a^2+b^2}}{2}\right]$$